

Causal Contrastive RL

This Week's Goal

Expand the occupancy measure formula and add bounds at each step

The resulting bound should satisfy three properties:

- ▶ **(P1) Theoretically sound:**
 - ▶ γ -contraction $\Rightarrow$ unique fixed point
 - ▶ Manski bound + monotonicity $\Rightarrow$ valid lower bound
- ▶ **(P2) Tighter than the naive "trajectory-level" Manski bound**

Step 1: Occupancy Measure $\rightarrow$ Bellman Recursion

Original definition (infinite sum, no per-step transition exposed):

$$d^\pi(s_g | s, a) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p_t^\pi(s_t = s_g \mid s_0 = s, a_0 = a)$$

Bellman recursion (per-step transition p now visible):

$$d^\pi(s_g | s, a) = (1 - \gamma) \mathbb{I}[s = s_g] + \gamma \sum_{s'} p(s' | s, a) \sum_{a'} \pi(a' | s') d^\pi(s_g | s', a')$$

Key point: After rewriting, the per-step transition $p(s' | s, a)$ appears explicitly — this is the quantity affected by confounding, where Manski bound applies.

Proof: Occupancy Measure $\equiv$ Bellman Recursion

Claim: The following two expressions are **exactly equivalent**:

$$\begin{aligned}d^\pi(s_g|s, a) &= (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p_t^\pi(s_t=s_g \mid s_0=s, a_0=a) \\&= (1 - \gamma) \mathbb{P}[s=s_g] + \gamma \sum_{s'} p(s'|s, a) \sum_{a'} \pi(a'|s') d^\pi(s_g|s', a')\end{aligned}$$

Proof sketch:

1. **Separate $t=0$:** $d^\pi = (1-\gamma) \underbrace{p_0^\pi(s_0=s_g|s_0=s)}_{=\mathbb{P}[s=s_g]} + (1-\gamma) \sum_{t=1}^{\infty} \gamma^t p_t^\pi$

2. **Time-shift ($k = t-1$):**

$$= (1-\gamma) \mathbb{P}[s=s_g] + \gamma (1-\gamma) \sum_{k=0}^{\infty} \gamma^k p_{k+1}^\pi(s_{k+1}=s_g)$$

3. **Law of total expectation** (condition on $S_1=s', A_1=a'$) + **Markov property**:

$$p_{k+1}^\pi(s_{k+1}=s_g|s, a) = \sum_{s'} p(s'|s, a) \sum_{a'} \pi(a'|s') p_k^\pi(s_k=s_g|s', a')$$

4. **Recognize the definition:**

$$(1-\gamma) \sum_{k=0}^{\infty} \gamma^k p_k^\pi(s_k=s_g|s', a') = d^\pi(s_g|s', a')$$



Step 2: Per-Step Manski Bound $\rightarrow$ Causal Bellman Equation

Decompose by behavior policy's natural choice:

- ▶ Naturally chose a (proportion $\rho(a|s)$): consistency axiom $\Rightarrow$ use p_e
- ▶ Did not choose a (proportion $1 - \rho(a|s)$): counterfactual $\Rightarrow$ worst case

Causal Bellman Equation:

$$\begin{aligned}\underline{d}^\pi(s_g|s, a) &= (1 - \gamma)\mathbb{1}[s = s_g] \\ &\quad + \gamma \rho(a|s) \sum_{s'} p_e(s'|s, a) \sum_{a'} \pi(a'|s') \underline{d}^\pi(s_g|s', a') \\ &\quad + \gamma (1 - \rho(a|s)) \min_{s' \in \mathcal{S}} \sum_{a'} \pi(a'|s') \underline{d}^\pi(s_g|s', a')\end{aligned}$$

where $V_{\underline{d}}(s') := \sum_{a'} \pi(a'|s') \underline{d}^\pi(s_g|s', a')$ is the policy-averaged occupancy.

(P1) Theoretical Properties ✓

Operator definition:

$$(\underline{T}f)(s, a) = (1-\gamma)\mathbb{I}[s = s_g] + \gamma\rho \sum_{s'} p_e V_f(s') + \gamma(1-\rho) \min_{s' \in S} V_f(s')$$

γ -contraction in sup-norm:

$$\|\underline{T}f - \underline{T}g\|_{\infty} \leq \gamma \|f - g\|_{\infty}$$

(Weighted sum and min operator both non-expansive.)

Unique fixed point: Banach fixed point theorem.

Valid lower bound: Manski bound gives $d^{\pi} \geq \underline{T}d^{\pi}$. Apply monotonicity iteratively:

$$d^{\pi} \geq \underline{T}d^{\pi} \geq \underline{T}^2 d^{\pi} \geq \dots \rightarrow \underline{d}^{\pi}$$

(P2) Tighter than Trajectory-Level Bound?

Key insight: Tightness depends on whether $\min_{s'} V_{\underline{d}}(s') > 0$ or $= 0$.

Experiment on Windy Corridor (with lava):

	Valid ($\underline{d}^\pi \leq d^\pi$)?	Tighter than $\rho \cdot d_e^\pi$?
Result	✓	100% loosened

Why: DEAD state $\Rightarrow V_{\underline{d}}(\text{DEAD}) = 0 \Rightarrow \min_{s'} V_{\underline{d}}(s') = 0$ always $\Rightarrow$ the $(1-\rho)$ term contributes nothing.

Example: state (12,5,R), one step from goal:

d_{true}^π	$\rho \cdot d_e^\pi$	$\underline{d}^\pi$
0.89	0.74	0.08

(Q3) Open Questions for Discussion

Question 1: Over-conservatism

Question 2: Objective Mismatch